Numerical Analysis - Exercise Sheet 1

Course: Introduction to Numerical Analysis (NUM201)
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Summary

Exercise sheet with short tasks for error analysis, bisection, Newton method, and interpolation.

Exercise 1 - Error classification

- For each situation, identify whether the main problem is round-off, truncation, or modeling error.

- Explain the reason in one sentence for each item.

Exercise 2 - Bisection method

- Use the bisection method on f(x) = x^3 - x - 2 over the interval [1, 2].

- Perform four iterations and write the midpoint after each iteration.

Exercise 3 - Newton method

- Starting from x0 = 1.5, apply two Newton iterations to f(x) = x^2 - 2.

- Compare the result with the exact square root of 2.

Exercise 4 - Interpolation

- Given points (0,1), (1,3), and (2,2), build a linear interpolation between each neighboring pair.

- Estimate the value at x = 1.5.

Quick reference

Task	Expected submission
1	Short written explanation
2	Iteration table
3	Two computed values
4	Interpolation formula and estimate